

Exam Electrodynamics Final

Neighbours

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SEMESTER: WS 2024

DATE: 24.01.2025

TIME: 10:45 - 12:15

First and Last Name

Student Registration Number

Grading Scheme	$\geq 90\%$	1
	$\leq 80\%$ and $\geq 90\%$	2
	$\leq 70\%$ and $\geq 80\%$	3
	$\leq 60\%$ and $\geq 70\%$	4
	$\leq 60\%$	5

Result:

___/ max. 101 points

Grade:

Student Cohort MA-MECH-24-VZ-ET

Study Programme M.Sc Smart Technologies

Permitted Tools Two hand-written sheets of paper are allowed.

Important Notes

Unnecessary Items

Place all items not relevant to the test (including mobile phones, smartwatches, etc.) out of your reach.

Identification (ID)

Lay your student ID or an official ID visibly on the table in front of you.

Examination Sheets

Use only the provided examination sheets and label each sheet with your name and your student registration number. The sheets be labelled on the front. Do not tear up the examination sheets.

Writing materials

Do not use a pencil or red pen and write legibly.

Please read the following instructions carefully.

- You have **90 minutes** to complete this exam. This question booklet contains 4 question(s), 6 pages (including the cover) for the total of 101 points.
- Check to see if any pages are missing.
- All the questions are **compulsory** and all the notations used in the questions have their usual meaning taught at the lectures and done in practice.
- **Read the instructions for individual questions carefully** before answering the questions.

Question	Maximum Point	Received Point
Line Integrals	20	
Electric Fields	30	
Gaussian Sphere	20	
Magnetic Field of a Straight Wire	30	
Sum	101	

[Q1] Line Integrals _____ 20

Find the circulation of the field

$$\mathbf{F} = (x - y) \hat{x} + x \hat{y}$$

around the circle

$$\boldsymbol{\ell}(t) = (\cos t) \hat{x} + (\sin t) \hat{y} + (0) \hat{z}$$

with a range of $0 \leq t \leq 2\pi$.

[Q2] Electric Fields _____ 30

One of these is an **impossible electrostatic field**. Please check each case.

a. $\mathbf{E}_1 = (y^2) \hat{x} + (2xy + z^2) \hat{y} + (2yz) \hat{z} \text{ V m}^{-1}$, (15)

b. $\mathbf{E}_2 = (xy) \hat{x} + (2yz) \hat{y} + (3xz) \hat{z} \text{ V m}^{-1}$. (15)

[Q3] Gaussian Sphere _____ 20

Find the field outside a uniformly charged solid sphere of radius R and total charge q . (20)

[Q4] Magnetic Field of a Straight Wire 30

Find the magnetic field a distance s from a long straight wire carrying a steady current I . (30)

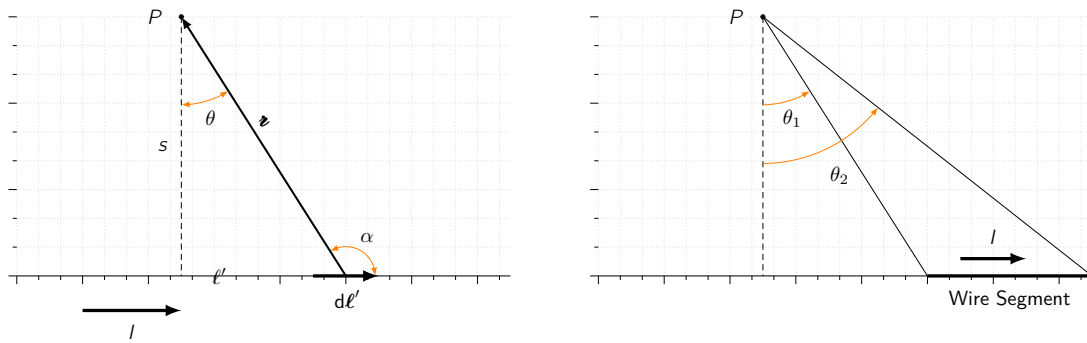


Figure 1: Diagrams to be use in question "Magnetic Field of a Straight Wire".

Formula Sheet

Vector Derivatives

Cartesian

Line Element	$d\mathbf{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$
Volume Element	$d\tau = dx dy dz$
Gradient	$\nabla t = \frac{\partial t}{\partial x} \hat{x} + \frac{\partial t}{\partial y} \hat{y} + \frac{\partial t}{\partial z} \hat{z}$
Divergence	$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$
Curl	$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z}$
Laplacian	$\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$

Cylindrical

Line Element	$d\mathbf{l} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$
Volume Element	$d\tau = s ds d\phi dz$
Gradient	$\nabla t = \frac{\partial t}{\partial s} \hat{s} + \frac{1}{s} \frac{\partial t}{\partial \phi} \hat{\phi} + \frac{\partial t}{\partial z} \hat{z}$
Divergence	$\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$
Curl	$\nabla \times \mathbf{v} = \left(\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right) \hat{s} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right) \hat{\phi} + \frac{1}{s} \left[\frac{\partial}{\partial s} (s v_\phi) - \frac{\partial v_s}{\partial \phi} \right] \hat{z}$
Laplacian	$\nabla^2 t = \frac{\partial^2 t}{\partial s^2} + \frac{\partial^2 t}{\partial \phi^2} + \frac{\partial^2 t}{\partial z^2}$

Fundamental Constants

$\epsilon_0 = 8,85 \cdot 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$	Permittivity of free space
$\mu_0 = 4\pi 1 \cdot 10^{-7} \text{ N A}^{-2}$	Permeability of free space
$c = 3 \cdot 10^8 \text{ m s}^{-1}$	Approximate speed of light
$e = 1,6 \cdot 10^{-19} \text{ C}$	Charge of electron
$m_e = 9,11 \cdot 10^{-31} \text{ kg}$	Mass of the electron

Spherical

Line Element	$d\mathbf{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$
Volume Element	$d\tau = r^2 \sin \theta dr d\theta d\phi$
Gradient	$\nabla t = \frac{\partial t}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial t}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial t}{\partial \phi} \hat{\phi}$
Divergence	$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$
Curl	$\nabla \times \mathbf{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\phi}{\partial \phi} \right] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\phi}$
Laplacian	$\nabla^2 t = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial t}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial t}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 t}{\partial \phi^2}$

Vector Identities

Triple Products

- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$
- $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B} (\mathbf{A} \cdot \mathbf{C}) - \mathbf{C} (\mathbf{A} \cdot \mathbf{B})$

Product Rules

- $\nabla (fg) = f (\nabla g) + g (\nabla f)$
- $\nabla (\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A}$
- $\nabla \cdot (f \mathbf{A}) = f (\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
- $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
- $\nabla \times (f \mathbf{A}) = f (\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
- $\nabla \times \mathbf{A} \times \mathbf{B} = (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A} (\nabla \cdot \mathbf{B}) - \mathbf{B} (\nabla \cdot \mathbf{A})$

Second Derivatives

- $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ (i.e., divergence of a curl is **always** zero.)
- $\nabla \times (\nabla f) = 0$
- $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$

Fundamental Theorems

Gradient Theorem: $\int_a^b (\nabla f) \cdot d\mathbf{l} = f(b) - f(a)$

Divergence Theorem: $\int (\nabla \cdot \mathbf{A}) d\tau = \oint \mathbf{A} \cdot d\mathbf{a}$

Curl Theorem: $\int (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$

Coordinate Conversion

Spherical

$$x = r \sin \theta \cos \phi \quad r = \sqrt{x^2 + y^2 + z^2}$$

$$y = r \sin \theta \sin \phi \quad \theta = \tan^{-1} \left(\sqrt{x^2 + y^2} / z \right)$$

$$z = r \cos \theta \quad \phi = \tan^{-1} (y/x)$$

$$\hat{x} = \sin \theta \cos \phi \hat{r} + \cos \theta \cos \phi \hat{\theta} - \sin \phi \hat{\phi}$$

$$\hat{y} = \sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} + \cos \phi \hat{\phi}$$

$$\hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

Cylindrical

$$x = s \cos \phi \quad s = \sqrt{x^2 + y^2}$$

$$y = s \sin \phi \quad \phi = \tan^{-1} (y/x)$$

$$z = z \quad z = z$$

$$\hat{x} = \cos \phi \hat{s} - \sin \phi \hat{\phi}$$

$$\hat{y} = \sin \phi \hat{s} + \cos \phi \hat{\phi}$$

$$\hat{z} = \hat{z}$$

$$\hat{s} = \cos \phi \hat{x} + \sin \phi \hat{y}$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

$$\hat{z} = \hat{z}$$

Curl Operation

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Useful Physical Identities

Energy $U = \frac{1}{2} \int \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) d\tau$

Momentum $\mathbf{P} = \epsilon_0 \int (\mathbf{E} \times \mathbf{B}) d\tau$

Poynting Vector $\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B})$

Lorentz Force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

Potential $\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$ and $\mathbf{B} = \nabla \times \mathbf{A}$

Coulombs Law $\mathbf{F} = \frac{qQ}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2}$

Maxwell's Equations

Main Equations

In General: $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

In Material: $\nabla \cdot \mathbf{D} = \rho_f$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

Auxiliary Fields

Definition: $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$$

Linear Media: $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$

$$\mathbf{M} = \chi_m \mathbf{H}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{H} = \frac{1}{\mu} \mathbf{B}$$